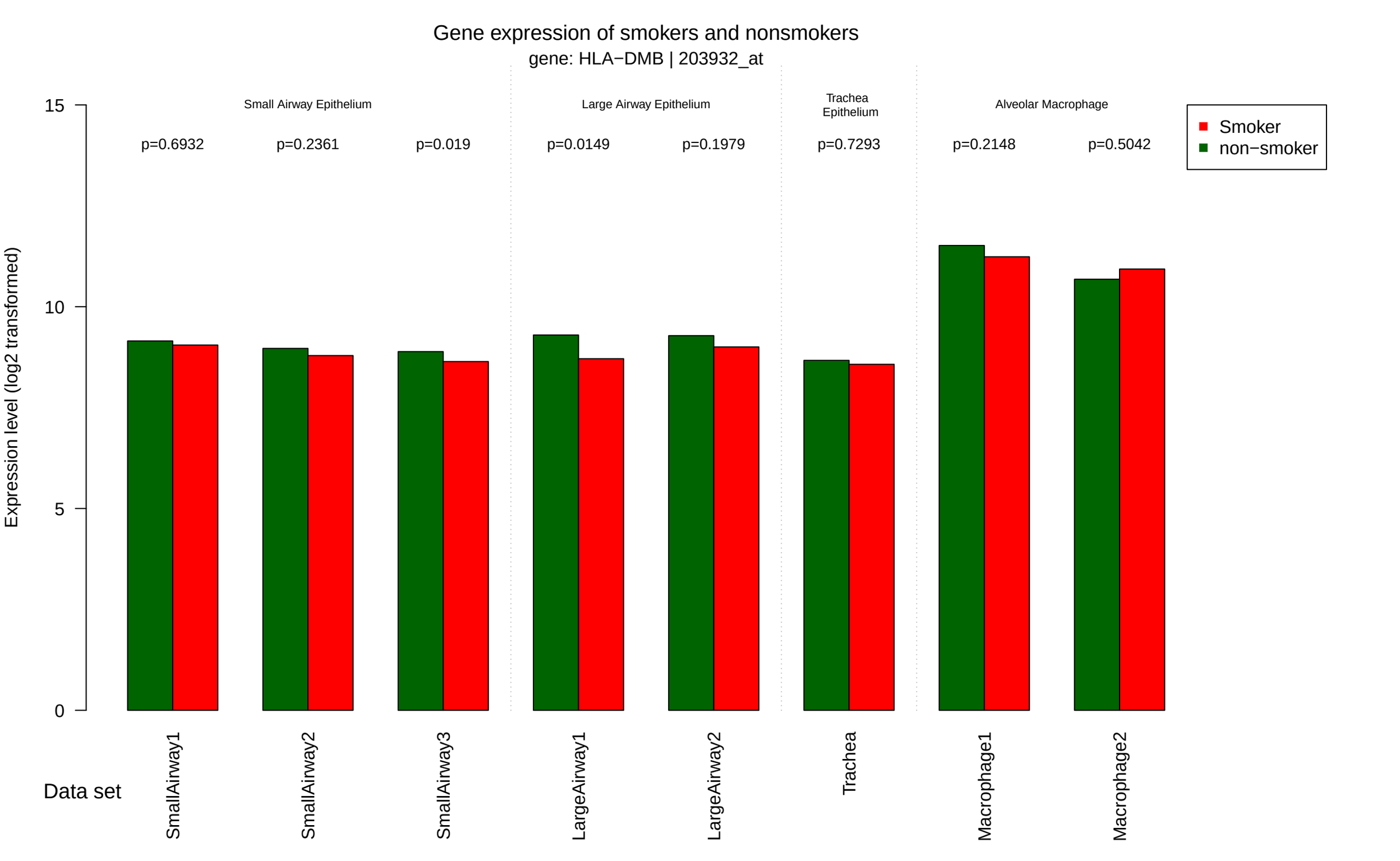
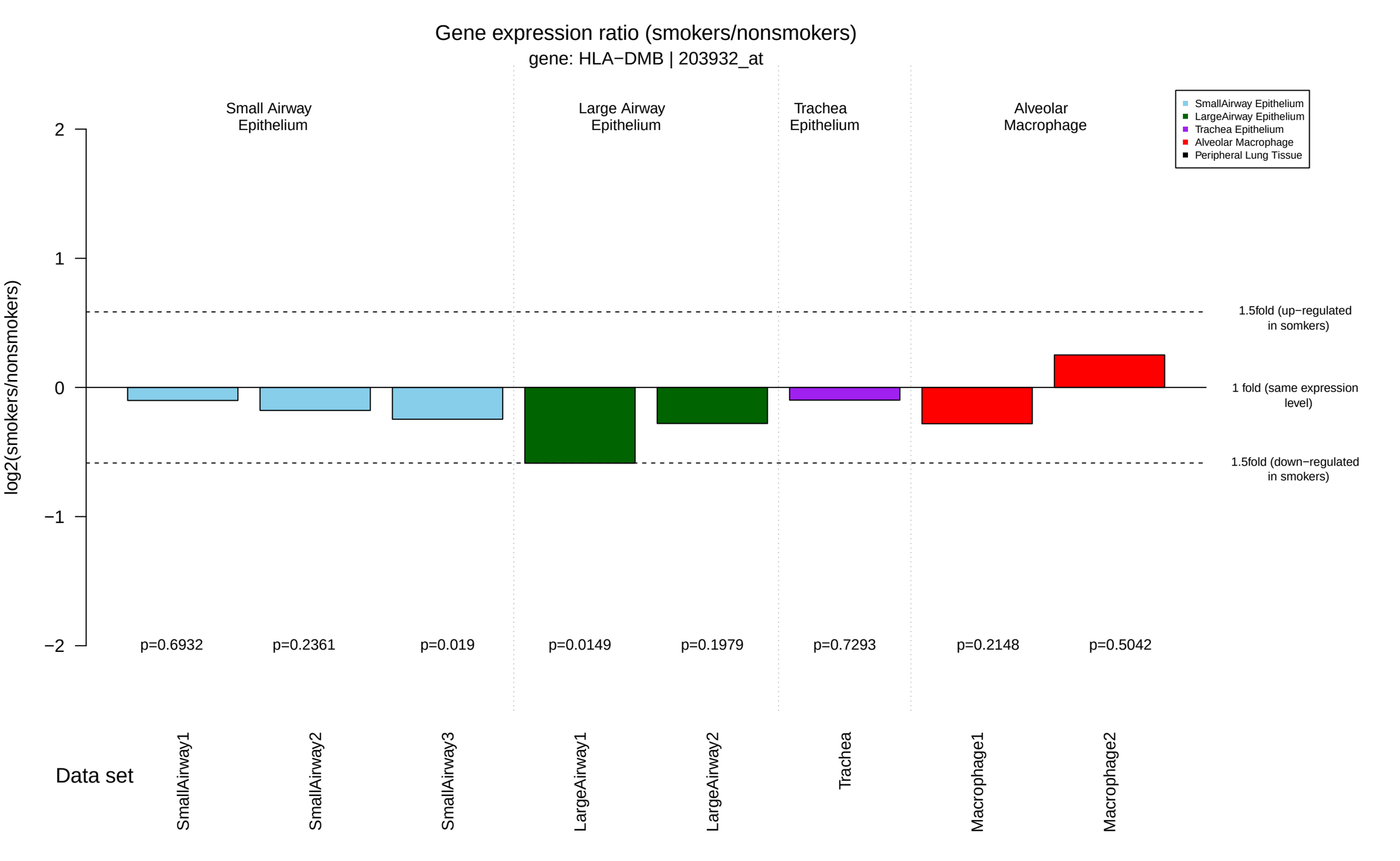
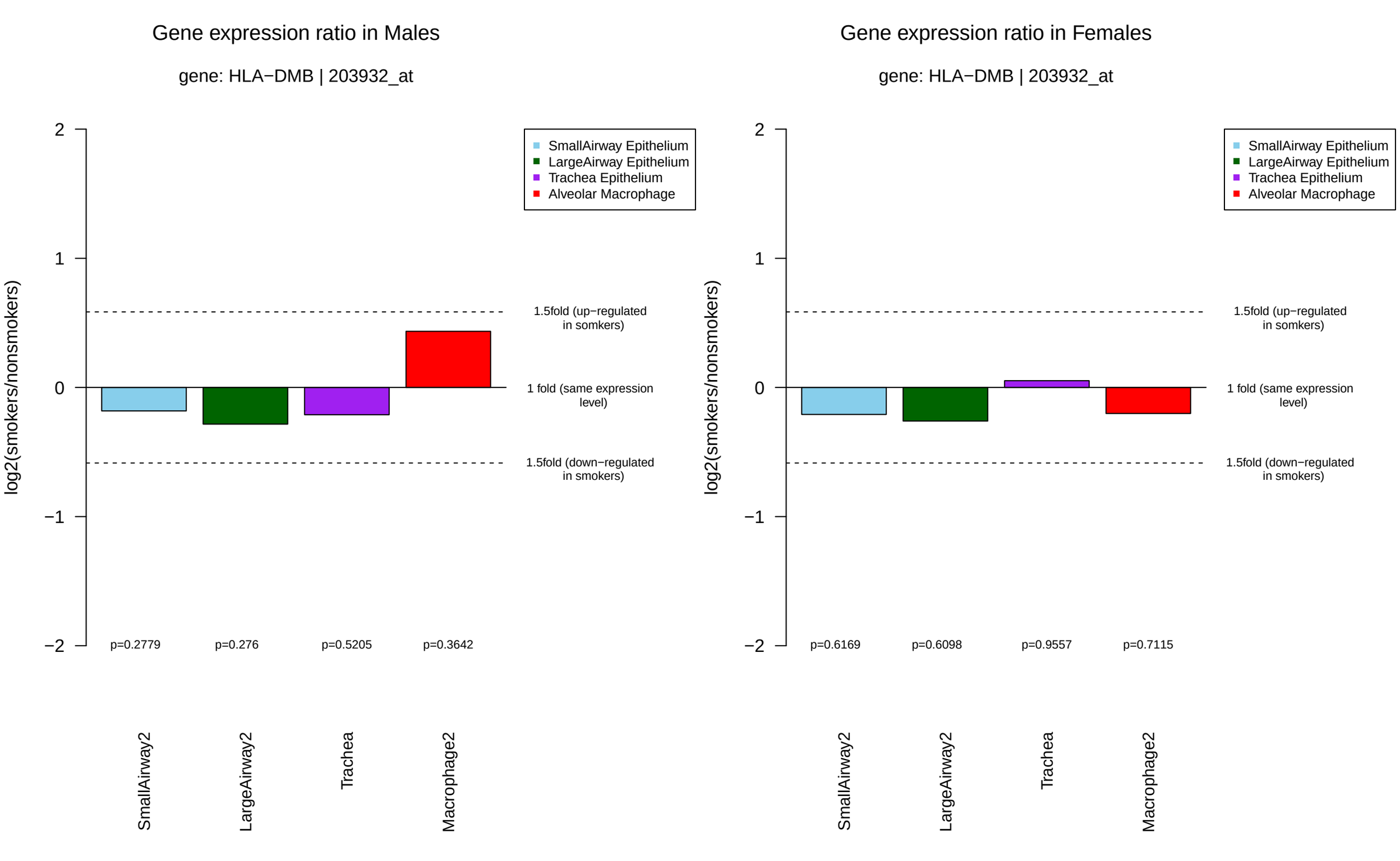
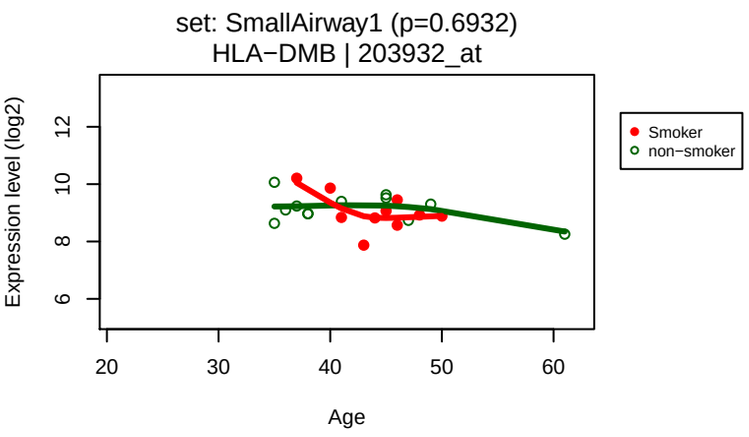
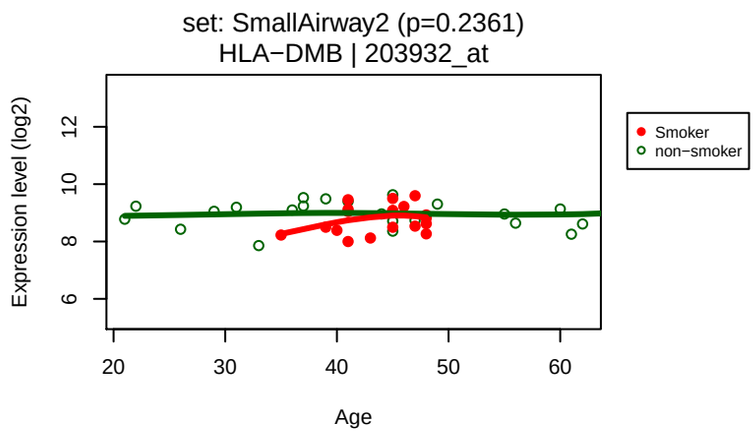
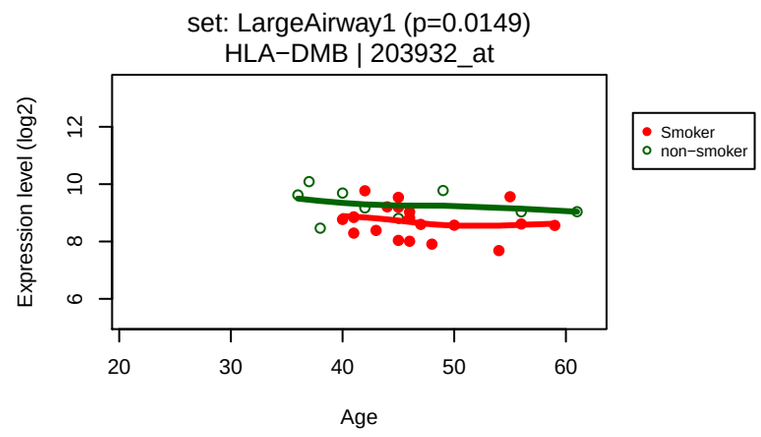

Gene expression of smokers and nonsmokers
gene: HLA−DMB | 203932_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6932
p=0.2361
p=0.019
p=0.0149
p=0.1979
p=0.7293
p=0.2148
p=0.5042
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

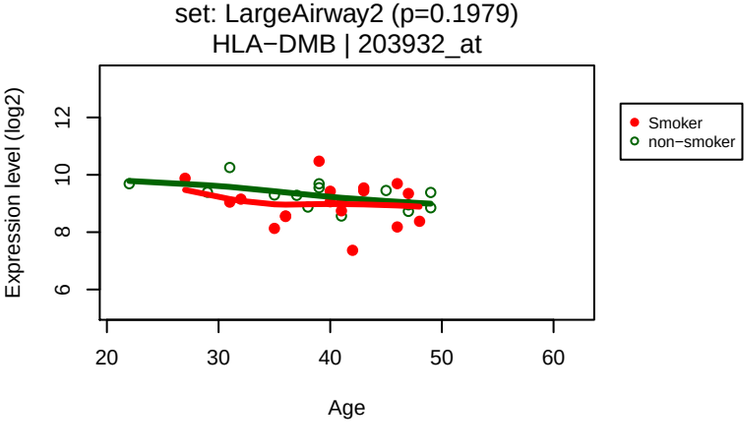

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DMB | 203932_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6932
p=0.2361
p=0.019
p=0.0149
p=0.1979
p=0.7293
p=0.2148
p=0.5042
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

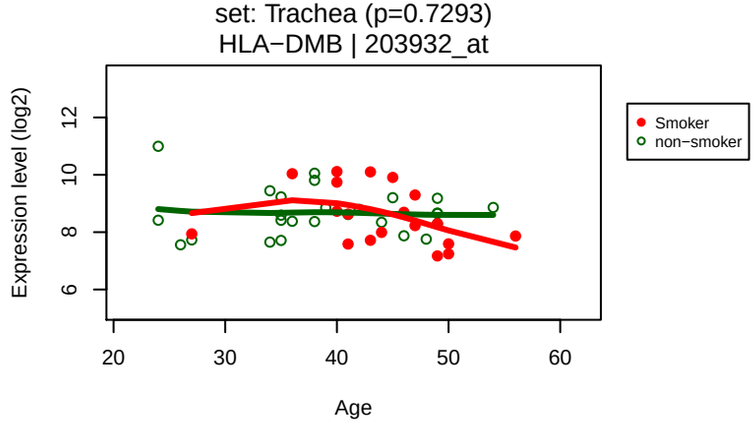

Gene expression ratio in Males
gene: HLA−DMB | 203932_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2779
p=0.276
p=0.5205
p=0.3642
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DMB | 203932_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6169
p=0.6098
p=0.9557
p=0.7115
SmallAirway2
LargeAirway2
Trachea
Macrophage2

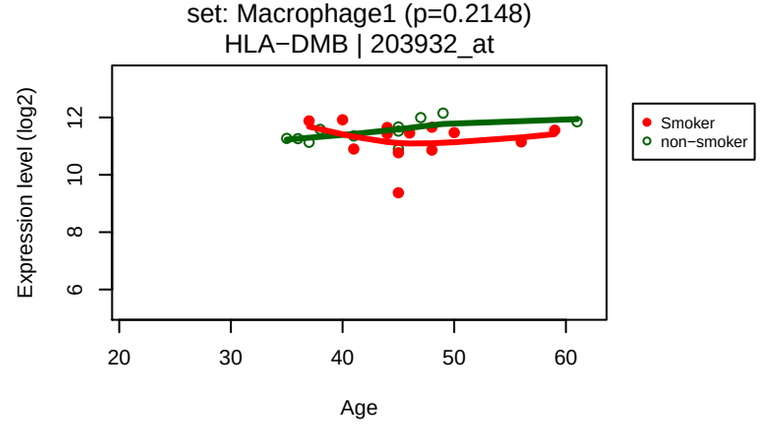

set: SmallAirway1 (p=0.6932)
HLA−DMB | 203932_at
Expression level (log2)
Age
Smoker
non−smoker

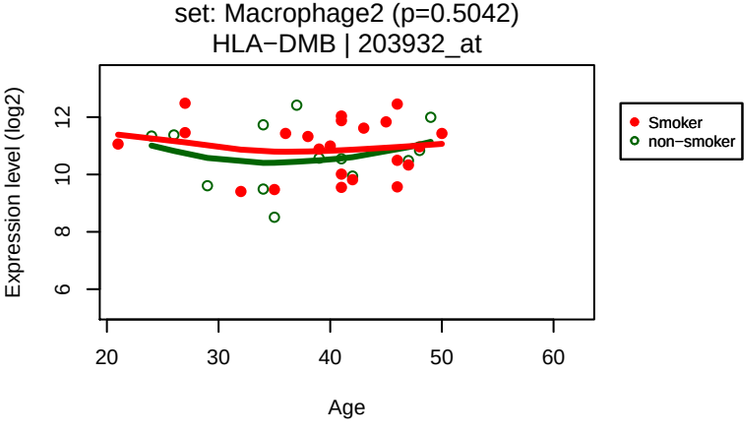

set: SmallAirway2 (p=0.2361)
HLA−DMB | 203932_at
Expression level (log2)
Age
Smoker
non−smoker

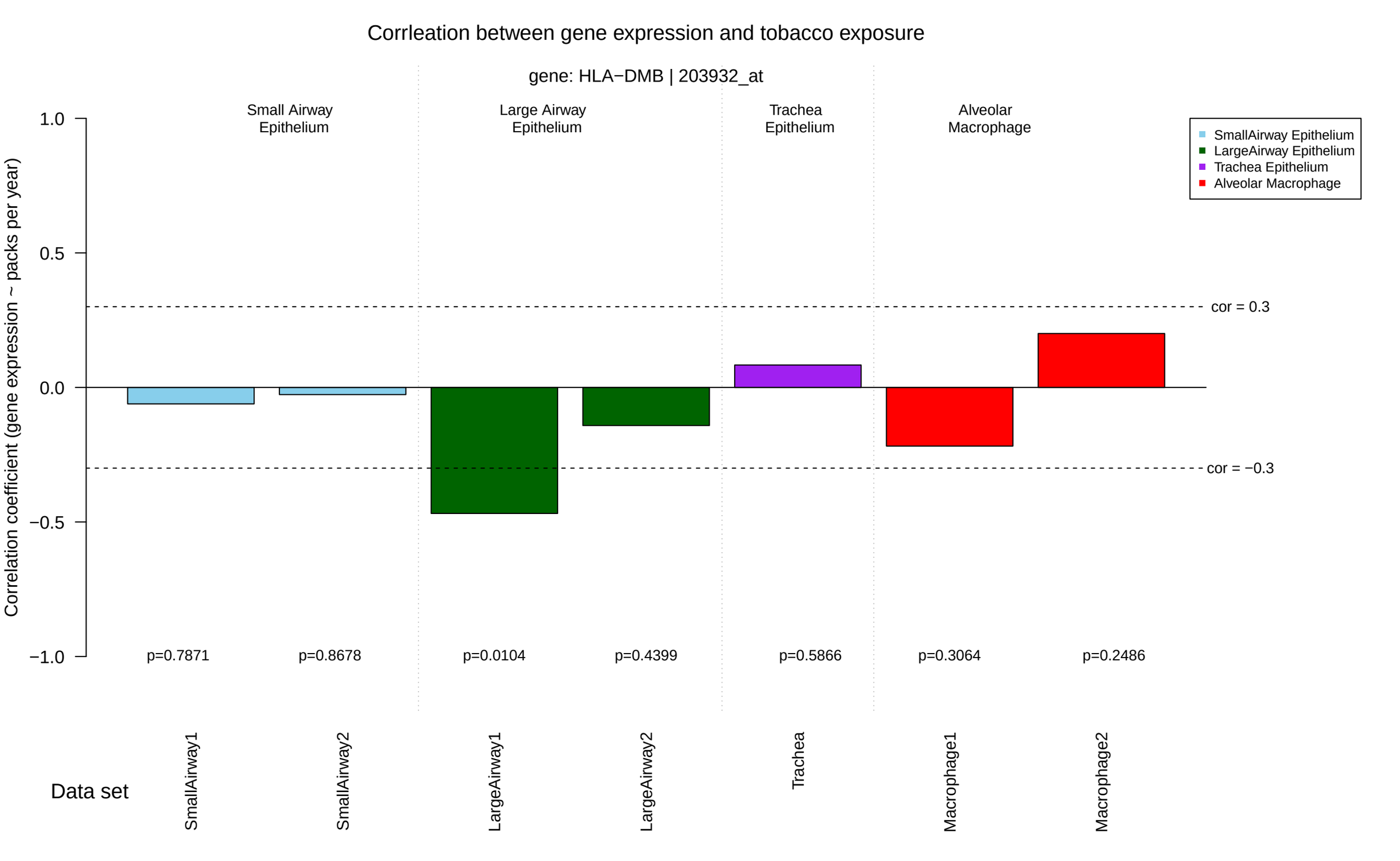

set: LargeAirway1 (p=0.0149)
HLA−DMB | 203932_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1979)
HLA−DMB | 203932_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7293)
HLA−DMB | 203932_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2148)
HLA−DMB | 203932_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5042)
HLA−DMB | 203932_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DMB | 203932_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.7871
p=0.8678
p=0.0104
p=0.4399
p=0.5866
p=0.3064
p=0.2486
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2